

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4.	(a)	(i)		Sodium / Na		1		1		
		(ii)	I	10-6 (1) / 10×100 Allow 6-10 = 40% (1)		2		2	2	
			II	More acidic		1		1		
			III	(pH) decrease			1	1		
		(iii)		Identification (1) Data (1) Judgement - can only be given if either identification or data mark given (1) e.g. for Mg Identification of Mg (1) 15.3, 15.4, 15.5 (1) Which are close together so agree / which are different in so disagree (1) e.g. for Ca Identification of Ca (1) 50.4, 50.1, 50.1 (1) Which are close together so agree / which are different in so disagree / only the same in two streams so disagree (1) Ignore reference to other ions			3	3		

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					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		100	1			1		1
		(ii)		Most of the {water vapour / steam} leaving flask A is {not collected / lost} (1) In B the steam changes back into {water / liquid } (using a condenser) (1) (so disagree)			2	2		3
		(iii)		too costly / requires a lot of energy / environmental impact / must be near the coast	1			1		1
				Question 4 total	2	4	6	12	2	4